

UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI

ICT LAB

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**A HYBRID TEMPLATE-BASED AND DEEP-LEARNING
FRAMEWORK FOR RNA THREE-DIMENSIONAL
STRUCTURE PREDICTION WITH
TEMPORAL LEAKAGE CONTROL AND
Geometry Refinement**

MASTER THESIS

DATA SCIENCE

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Declaration

[TO COMPLETE: Insert the declaration required by the university template.]

Acknowledgements

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Abstract

Predicting the three-dimensional structure of RNA from sequence remains difficult because RNA can adopt multiple conformations and experimental structures remain limited [1, 11, 10]. This thesis starts from the public method of the winning Stanford RNA 3D Folding team: retrieve related structures, transfer coordinates to the query, and supplement template models with a pretrained predictor. It then develops an auditable pipeline with time-safe TBM, two DRfold2 candidates, and Geometry Refinement. Five structures are produced without access to the native structure, matching the competition’s best-of-five evaluation [12, 27, 28].

The evidence is separated into local component studies and external hidden-set evaluation. The local protocol uses 60 temporally earlier RNAs to estimate geometric priors, 20 RNAs to select the configuration, and 20 family-disjoint newer RNAs for a single confirmatory evaluation. In the TBM component study, MMseqs2 returned a template for 8 of 20 calibration targets, whereas composite search produced candidates for all 20. The composite score therefore increased retrieval coverage, although its full weighted form did not clearly outperform global alignment alone.

On the 20 held-out RNAs, three TBM candidates reached a mean best-available TM-score of 0.565, whereas two DRfold2 candidates reached 0.503. Keeping all five candidates raised the score to 0.588, an absolute gain of 0.023 with a 95% confidence interval of [0.006, 0.048]. DRfold2 supplied the stronger candidate on 8 of 20 targets, indicating complementarity rather than higher average accuracy. Geometry Refinement increased C1'-IDDT from 0.660 to 0.666 and reduced SW-RMSD9 from 2.651 to 2.611 Å, while TM-score changed from 0.588 to 0.587. The method therefore improved short-range accuracy without increasing global-fold accuracy.

On Kaggle, the thesis TBM pipeline scored 0.601 public and 0.602 private, while the final pipeline scored 0.628 and 0.614. The final private score was 0.021 above John’s public TBM-only notebook, 0.038 above his public hybrid notebook, and 0.036 above the winning team’s 0.578 leaderboard score. These are pipeline-level comparisons because the systems differ in data, retrieval, pretrained sources, and postprocessing. Controlled local

experiments identify candidate diversity as the main mechanism behind the TM-score gain, whereas Geometry Refinement provides a separate short-range benefit.

Keywords: RNA three-dimensional structure, template-based modeling, DRfold2, TM-score, C1'-IDDT, temporal safety, Geometry Refinement.

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Chapter 1

Introduction

1.1 Background and motivation

RNA regulates gene expression, catalyzes reactions, recognizes molecular partners, and forms ribonucleoprotein complexes. These functions depend not only on the order of A, C, G, and U, but also on the base pairs, motifs, and three-dimensional arrangements formed by the chain [2, 3, 4]. Secondary structure describes base pairing and local motifs, whereas tertiary structure captures how these elements interact in space [3, 4]. An RNA 3D predictor must consequently recover both the global fold and its local geometry.

X-ray crystallography, nuclear magnetic resonance, and cryogenic electron microscopy are major routes to experimental RNA structures. Each technique has a distinct domain of applicability, and cryo-EM has expanded access to large RNA-containing complexes; nevertheless, sample preparation, data acquisition, and model reconstruction remain resource-intensive [1, 5]. The PDB provides the global archive for experimentally determined macromolecular structures, but RNA remains less represented than protein and many RNA families lack close structural templates [6, 1]. This gap motivates computational prediction from sequence.

Current RNA 3D methods draw on physical principles, knowledge extracted from observed structures, and learned patterns. Template-based modeling can be highly accurate when a suitable structural relative is recognized, whereas deep models can produce candidates when direct template retrieval is weak [11, 16, 17, 18]. Blind RNA-Puzzles and CASP assessments show that rankings vary across targets, lengths, fold novelty, and permitted information [7, 9, 10]. A bank of candidates from different sources is therefore attractive when the evaluation retains the best of several predictions.

The Stanford RNA 3D Folding competition provides a particularly relevant setting.

More than 1,700 teams predicted 43 previously unreleased structures, and both leading systems used TBM. Post-competition analysis further showed that combining TBM and deep learning could outperform individual components on the same test set [12]. This observation does not make every RNA a template-rich problem. Instead, it motivates two questions: how to retrieve structures without future information, and how to cover targets for which templates are inadequate.

1.2 Research problem

A TBM pipeline can receive an artificially high score if its library contains a structure released after the prediction cutoff. This temporal leakage uses information that is public at rerun time but unavailable in the original blind setting. RNA-Puzzles and CASP explicitly separate prediction from unreleased structures, making the data cutoff part of the scientific protocol [7, 9, 12]. This thesis requires every template to predate its target and excludes the target’s own reference PDB entries.

Even under a valid cutoff, a single retrieval system may miss useful templates. MM-seqs2 supports fast, sensitive sequence search at database scale [13], but structural relatives do not always present a sufficiently strong homolog for the fast retrieval path. We therefore complement MMseqs2 with a search heuristic based on global alignment, local alignment, sequence composition, and 3-mer overlap. Its purpose is to increase candidate recall rather than define an optimal biological score.

TBM cannot provide a strong model for every target. DRfold2 is therefore introduced as an independent source rather than a replacement for templates. DRfold2 extends the line of work that combines end-to-end learning with learned geometric potentials for RNA 3D prediction [16, 18]. Retaining three TBM and two DRfold2 models preserves template strength on easier targets while reserving two slots for folds that the deep model may recover better.

Candidates from both branches may still contain irregular backbone distances, clashes, or local distortions. Previous nucleic-acid refinement tools show that better geometric quality does not guarantee systematic improvement in native RMSD [20, 21]. The refinement objective in this thesis is therefore narrow: improve local C1’ geometry while keeping TM-score within a predeclared preservation margin. Refinement is not presented as a second global-fold predictor.

1.3 Research questions

The thesis addresses three questions in the order in which information enters the pipeline:

- RQ1.** What does composite search add to MMseqs2 in a time-safe TBM pipeline, and which ranking components produce measurable changes?
- RQ2.** When candidate count is controlled, does DRfold2 complement the targets missed by TBM?
- RQ3.** Does Geometry Refinement improve C1'-lDDT and sliding-window RMSD while preserving global TM-score?

These questions form a testable progression. RQ1 explains candidate availability, RQ2 tests source diversity, and RQ3 identifies the structural scale affected by postprocessing. The leaderboard is used as an external assessment of the frozen pipeline, not as a substitute for local ablation.

1.4 Contributions

The thesis does not present the full pipeline as an invention built from scratch. Its starting points are John's public TBM notebook, which is associated with the winning team, and his public hybrid notebook combining Boltz, DRfold2, and template fallback [28, 29]. The thesis changes data controls, retrieval, ranking, candidate allocation, and postprocessing, and evaluates these changes separately.

The first contribution is an auditable time-safe TBM branch that unifies fast and composite retrieval, enforces release-date and self-PDB filters, realigns candidates, ranks them by identity, coverage, and completeness, and transfers C1' coordinates. Component studies evaluate each decision instead of inferring its value from a single final score.

The second contribution is quantitative evidence of complementarity between TBM and DRfold2. The study reports each source, their union, and fixed-size candidate banks. Fixed-N comparisons partially separate genuine source effects from the mechanical benefit of adding more prediction slots.

The third contribution is conservative Geometry Refinement evaluated with a held-out protocol. Priors are estimated on 60 RNAs, the configuration is selected on 20, and the method is frozen before opening 20 newer RNAs. Raw candidates, a simple smoothing baseline, the full objective, cumulative ablation, and leave-one-out ablation are all reported.

Finally, the thesis provides a two-level explanation of leaderboard performance. The 0.614 private score establishes external value for the complete pipeline, while local experiments examine candidate complementarity and Geometry Refinement separately. This separation avoids assigning the full Kaggle difference to a component that has not yet been isolated on the private set.

1.5 Scope and organization

The system predicts one C1' coordinate per nucleotide and returns five structures per target. Confirmatory local analysis covers single-chain RNAs up to 96 nucleotides; Kaggle provides an external check on a different distribution that includes longer targets. Full-atom chemistry, ions, ligands, chemical modifications, and RNA-protein interfaces are outside the evaluated representation, although each can be important to RNA structure and function [4, 1].

Chapter 2 reviews the scientific and computational background. Chapter 3 describes the data and leakage controls. Chapter 4 presents the pipeline, and Chapter 5 defines the experiments. Results are reported in Chapter 6, followed by discussion in Chapter 7 and conclusions in Chapter 8.

Chapter 2

Background and Related Work

2.1 RNA structural organization

RNA is a polymer with a sugar-phosphate backbone and four primary bases. Watson-Crick interactions, base stacking, noncanonical pairs, ions, and tertiary contacts contribute to structural stability [3, 4]. Interactions that are distant in sequence can become adjacent in space, allowing a local motif error to alter the molecular arrangement. Local and global quality must therefore be measured with different metrics.

This thesis uses the C1' trace as a common representation. For a sequence of length L , a candidate is

$$X = [\mathbf{x}_1, \dots, \mathbf{x}_L]^T \in \mathbb{R}^{L \times 3}, \quad (2.1)$$

where $\mathbf{x}_i$ is the C1' coordinate of nucleotide i . This coarse-grained form matches the competition output [12, 27]. It does not encode base orientation or full chemistry, so conclusions are limited to fold and C1'-trace quality.

Rigid rotation and translation do not alter internal structure. Kabsch superposition finds the rotation that minimizes squared deviations for known correspondence [25]. IDDT avoids global superposition, whereas TM-score searches for a structural alignment and uses length-dependent normalization [22, 24, 23]. The metrics consequently respond to distinct scales of structural error.

2.2 Template-based modeling

TBM assumes that a known structure can provide a scaffold for a related query. A typical workflow retrieves templates, aligns query and template, transfers corresponding coordinates,

builds unsupported regions, and refines the model [11, 12]. Accuracy depends on both recognizing a suitable structure and aligning residues correctly. High identity over a short fragment may be less useful than weaker similarity over nearly the complete target.

Needleman-Wunsch solves global sequence alignment, whereas Smith-Waterman identifies the best local alignment [14, 15]. The two algorithms expose different signals: global alignment measures whole-chain agreement and local alignment recognizes conserved regions. The composite search combines both with composition and 3-mer features to broaden retrieval.

MMseqs2 enables sensitive search over large sequence collections [13]. Retrieval speed, however, is not equivalent to structural coverage. The experiments therefore report availability, conditional quality when a hit exists, and whole-cohort means separately. This prevents the 12 targets without MMseqs2 hits from silently disappearing from the evaluation.

2.3 Deep RNA structure prediction

Recent deep models learn sequence representations, geometric distributions, or coordinates for RNA. ARES applies geometric deep learning to structure scoring, DRfold combines end-to-end prediction with learned potentials, RhoFold+ uses an RNA language model, and DRfold2 develops a more efficient predictive pipeline [19, 16, 17, 18]. Although these approaches reduce direct dependence on a close template, their accuracy remains target and benchmark dependent [10, 18].

DRfold2 is not framed as a universal replacement for TBM in this thesis. The sources have distinct failure modes: TBM depends on retrieval and alignment, whereas a pretrained model depends on its training distribution and learned evolutionary signals. A common candidate bank exploits this difference, and the held-out experiment tests whether the difference is useful in practice.

2.4 Structure refinement

Refinement searches the neighborhood of a candidate for structures that better satisfy geometric or energetic constraints. QRNAS optimizes nucleic-acid models with an extended energy function, and Arena reconstructs full-atom RNA from coarse-grained predictions [20, 21]. This literature distinguishes stereochemical regularization from movement toward the native structure. Better bonds and fewer clashes do not guarantee a better global fold.

The refinement in this thesis follows the same distinction. A source restraint preserves the initial candidate; distance, angle, and torsion priors regularize the trace; and clash, kink, and radius-of-gyration terms target specific artifacts. Effectiveness is measured primarily with TM-score, C1'-IDDT, and sliding-window RMSD rather than the optimized loss itself [22, 24, 23].

2.5 Structural evaluation

TM-score was introduced as a length-normalized measure of structural similarity [22], and US-align extends structural alignment to nucleic acids and macromolecular complexes [23]. The competition scores each target using the best combination of five predictions and available native conformations, then averages over targets [12]. In this thesis, best-of-five is an evaluation oracle only; inference never uses native structures to select a candidate.

IDDT evaluates local distance preservation without global superposition [24]. We compute it on C1' atoms and denote it C1'-IDDT. Sliding-window RMSD superposes each contiguous window by Kabsch and then measures coordinate deviation; windows of 9 and 15 nucleotides respond more strongly to local distortions than a 31-residue window or whole-structure RMSD [25]. Together, these metrics separate local trace correction from global fold change.

2.6 Research gap

Benchmarking studies compare RNA 3D predictors, and the competition analysis shows that TBM remains competitive as structural databases grow [10, 12]. A leaderboard score, however, does not reveal which component produced the result, and a component study without a temporal cutoff can favor template search. The first gap is therefore a white-box study of TBM under a realistic release-date policy.

The second gap concerns source complementarity under controlled candidate count. A five-model union can beat a three-model bank merely because it has more opportunities. This thesis reports both union and fixed-N comparisons. The final gap is confirmatory evaluation of Geometry Refinement against raw candidates, a simple baseline, independent metrics, and ablations. These gaps map directly to the three research questions.

Chapter 3

Data and Evaluation Controls

3.1 Data sources and splits

Kaggle distributes two main table types. Sequence tables contain one row per RNA with a target identifier, nucleotide sequence, temporal cutoff, and system description. Label tables contain one row per nucleotide with a position identifier and C1' coordinates for one or more experimental conformations. This schema separates the input available to a predictor from the native structure used only for training or evaluation. Table 3.1 summarizes the frozen project snapshot.

Table 3.1: Principal Kaggle files and their roles in the pipeline.

File	Rows	Meaning of one row	Role
train_sequences.csv	844	One RNA and metadata	Initial public training set
train_sequences.v2.csv	5,135	One RNA of 10-4,417 nucleotides	Source for temporal cohort selection
train_labels.v2.csv	3,677,095	One nucleotide and C1' coordinates	Public reference structures
validation_sequences.csv	12	One 30-720 nt CASP15 RNA	Local development and reproduction
validation_labels.csv	2,515	One nucleotide, up to 40 conformations	TM-score and local metrics
test_sequences.csv	12	Mock table equal to validation	Notebook check before hidden rerun
sample_submission.csv	2,515	Identifier and five XYZ predictions	Required 18-column output schema

The repository test table is the CASP15-based mock set. During scoring, the notebook receives hidden-test sequences at runtime but never receives their native coordinates.

Sequences are therefore inference inputs, whereas native coordinates remain hidden until Kaggle scores the five predictions [12, 27]. The snapshot also contains 856 initial and 2,534 v2 MSA files. The final pipeline does not read these files directly; reporting them distinguishes available competition data from the inputs actually used by each branch.

The template library is built from public PDB structures. The PDB is the single global archive for macromolecular structural data and supplies chain, sequence, coordinate, and release metadata [6]. The frozen project snapshot contains 8,670 structure files, 23,869 RNA or RNA-DNA chains, and approximately 10.86 million residues. About 99.9% of residues contain a C1' coordinate. The MMseqs2 index has roughly 20,844 valid chains, while the deduplicated composite library has 7,155 unique sequences. These counts describe the project artifact rather than the current PDB.

The initial development set contains 12 CASP15 targets of 30 to 720 nucleotides. CASP is a blind assessment in which target structures are unreleased during prediction, and its RNA targets have an official independent assessment [9]. Because this cohort is small and intentionally difficult, it is used for development, leakage illustration, and deployment checks rather than as the only confirmatory evidence.

The principal study uses 100 RNAs released between January 2024 and March 2025. Every target has three blindly generated TBM candidates and two DRfold2 candidates. The data are split temporally into 60 train, 20 calibration, and 20 validation RNAs as shown in Table 3.2. An audit found no sequence group or family group crossing a split.

Table 3.2: Data split for Geometry Refinement and complementarity analysis.

Split	RNAs	Release-date range	Role
Train	60	2024-01-10 to 2024-12-04	Estimate geometric priors
Calibration	20	2024-12-11 to 2025-01-08	Select and freeze configuration
Validation	20	2025-01-15 to 2025-03-26	One confirmatory evaluation

The dominant feature of Table 3.2 is role separation. Train creates statistics, calibration determines hyperparameters, and validation measures the frozen method. No validation result is used to revise the configuration. This protocol cannot remove every distribution bias, but it prevents the same target from selecting and confirming the method.

Kaggle supplies the external evaluation layer. The competition requires five predictions per RNA, scores them with US-align on C1' atoms, and reports an aggregate hidden-set

score [12, 27]. We retain public and private scores, the execution manifest, and the public CSV hash. Because native coordinates and per-target scores remain unavailable, the leaderboard evaluates the complete pipeline rather than replacing local component ablation.

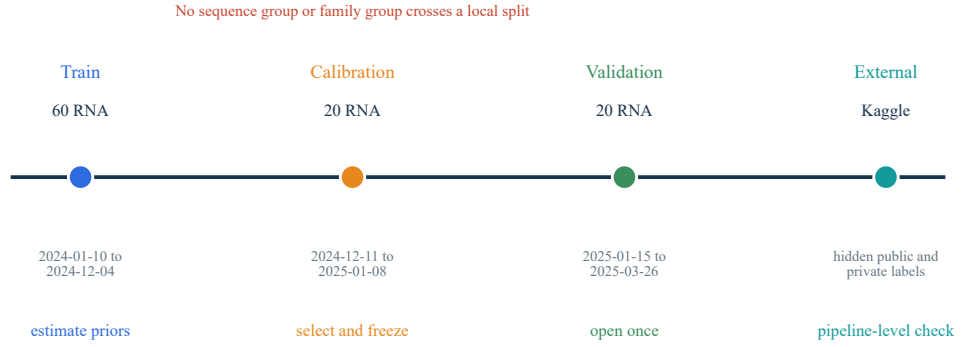


Figure 3.1: Two-level evidence protocol. Local splits are separated by time and family; Kaggle supplies an external hidden-set check of the complete pipeline.

3.2 Temporal and self-template exclusion

For target q with cutoff t_q , template s is valid only if

$$\text{release}(s) < t_q \quad \text{and} \quad \text{PDB}(s) \notin \mathcal{E}_q, \quad (3.1)$$

where $\mathcal{E}_q$ contains the target’s excluded PDB entries. A strict inequality avoids ambiguity for same-day releases. Missing cutoffs or release dates should fail closed rather than default to an admissible template.

Figure 3.2 uses the early CASP15 pipeline to illustrate this risk. Mean best-of-five TM was 0.161 with the temporal filter, 0.639 without it, and 0.957 when the target’s own template was deliberately allowed. These values predate composite search and are not directly comparable to the later 0.307 TBM result. Their sole purpose is to quantify the inflation caused by an invalid PDB snapshot [7, 9].

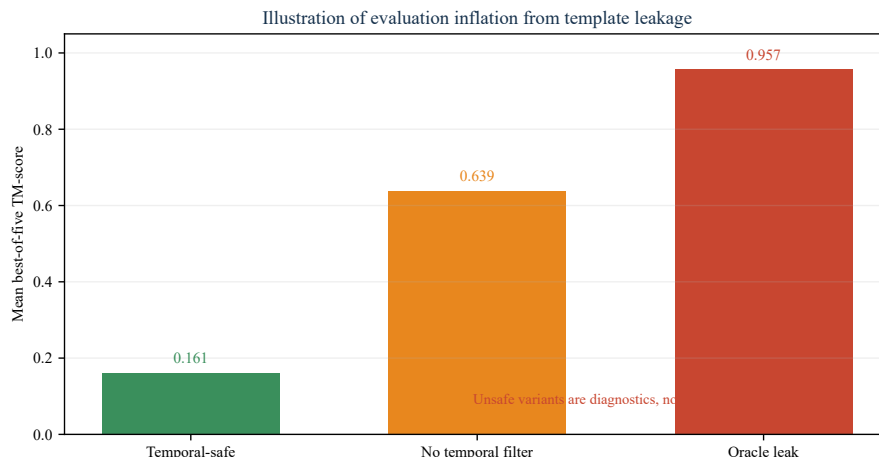


Figure 3.2: Evaluation inflation from template leakage on the development CASP15 cohort. The unsafe variants are diagnostics only.

3.3 Label-use policy

Native labels are read only by local evaluation scripts after candidate generation. Within the complementarity analysis, an “oracle” is a maximum used to measure the available ceiling of a candidate bank; it is never an inference-time selector. Geometry refinement receives only source coordinates, source confidence, and aggregate priors. For each candidate, the reference conformation used for local metrics is fixed from the raw candidate so that refinement cannot benefit by switching to an easier native.

An exact-sequence audit of CASP15 found two targets in the supplied DRfold2 training manifest: R1128 and R1138. Pretrained results on this cohort are therefore developmental and require a sensitivity analysis excluding both RNAs. The principal 100-RNA cohort is time- and family-separated with no sequence or family group crossing a split, although the thesis does not claim absence of every remote relationship to RNA language-model pretraining data.

The public execution manifest records `native_labels_used: false`; its mock output has 2,515 rows, 18 columns, correct identifiers, and finite coordinates. The public-file SHA-256 is

54fe6e6598a1c2924ef7cbf03482f7ecf3c2e52e8726a1609569f777b1ac8c9a

These checks establish schema and inference integrity on the 12 public RNAs. Kaggle produces the private score through a hidden rerun, so the 2,515-row file is not presented as the hidden output. Equation 3.1 governs template validity in both executions.

3.4 Statistical unit

The RNA target is the independent unit for every confidence interval. Candidates from one target share a sequence, native structure, and generation process, and are not treated as independent observations. For target effects d_i , we report

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i \quad (3.2)$$

and a percentile interval from 10,000 bootstrap samples of targets with replacement. Bootstrap resampling provides a nonparametric estimate of uncertainty from the observed sample [26]. The random seed is fixed at 2025.

Effects are oriented so that positive values indicate improvement. TM-score and lDDT use new minus baseline, whereas RMSD, clash, and deviation use baseline minus new. Tables also report medians and improve, tie, and regress counts when available, preventing a mean effect from hiding systematic failures.

Chapter 4

Proposed Pipeline

4.1 Overview

The pipeline accepts an RNA sequence and temporal cutoff, then generates candidates from two independent branches. TBM returns three ranked structures, and DRfold2 returns two structures by internal confidence. When DRfold2 exceeds its resource limit, TBM fills the unused positions. Geometry Refinement processes each candidate independently for 300 steps, and the writer emits exactly five C1' traces.

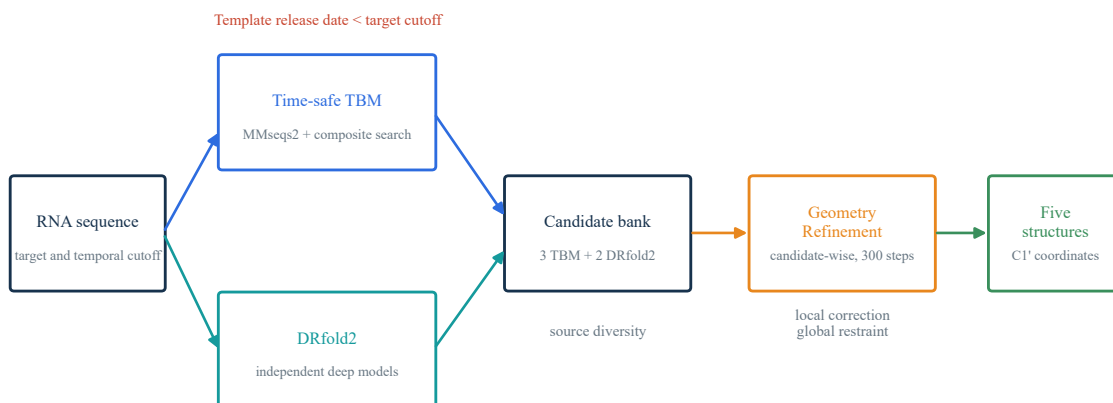


Figure 4.1: Final pipeline. Source diversity is created before candidate-wise refinement, with no native-guided selection or coordinate fusion.

The design follows two principles. First, diversity is introduced at the source level through three TBM and two DRfold2 candidates. Second, refinement remains close to each source to avoid disrupting an already correct fold. The two principles address distinct failure modes and are evaluated separately.

4.2 Time-safe TBM

4.2.1 Dual-path retrieval

The first path runs MMseqs2 over the template-sequence index. MMseqs2 uses accelerated filters and sensitive alignment to search large collections [13]. Raw hits are normalized to chain keys, identity, alignment coverage, and release dates. Hits that lack C1' coordinates or violate temporal validity are removed before ranking.

The second path scans the deduplicated library with

$$S_{\text{comp}}(q, s) = 0.4G(q, s) + 0.3L(q, s) + 0.2F(q, s) + 0.1K_3(q, s), \quad (4.1)$$

where G is global-alignment similarity, L is local-alignment similarity, F measures sequence-composition agreement, and K_3 is 3-mer overlap. The alignment terms use the Needleman-Wunsch and Smith-Waterman formulations [14, 15]. The weights are frozen heuristics, and their sensitivity is tested by ablation rather than assigned a probabilistic interpretation.

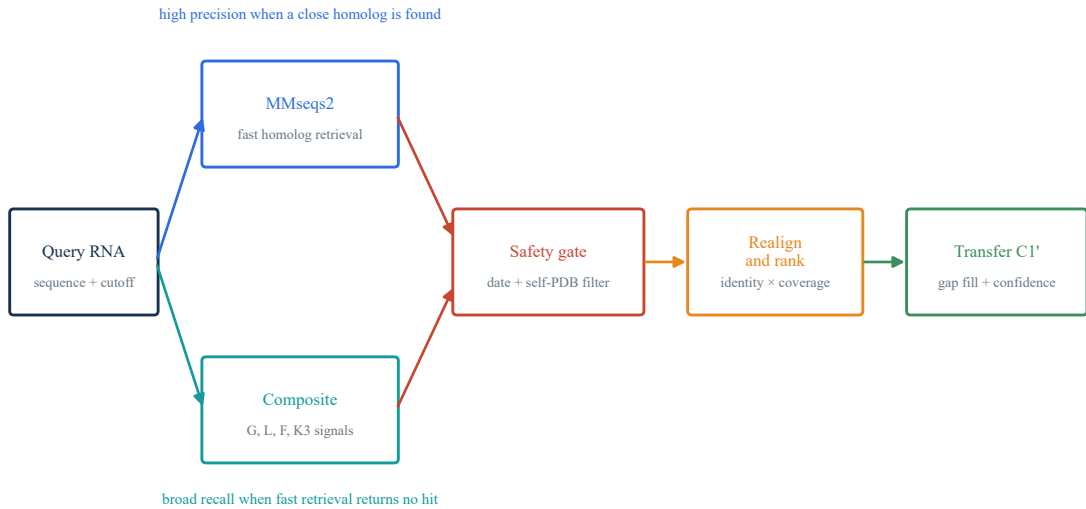


Figure 4.2: TBM flow. MMseqs2 prioritizes clear homologs, composite search broadens recall, and the safety gate precedes realignment and coordinate transfer.

The two lists are merged by chain key before temporal and self-PDB filtering. The design in Figure 4.2 keeps the roles separate: MMseqs2 is not considered inaccurate when no hit is returned, and composite search is not considered accurate merely because it always returns a candidate. Availability and conditional quality are reported separately.

4.2.2 Realignment and ranking

Every valid hit is globally realigned with match 2, mismatch -1 , gap opening -6 , gap extension -0.5 , and unpenalized terminal gaps. This final alignment establishes one correspondence for coordinate transfer. Let I denote identity on aligned positions, C_q the query fraction with supported coordinates, and C_s template completeness. The rank score is

$$S_{\text{rank}} = I C_q C_s. \quad (4.2)$$

The production pipeline sorts directly by the complete product IC_qC_s ; completeness is therefore a ranking component rather than a tie-breaker. For example, a 0.900-identity hit covering 0.400 of the query scores 0.360, whereas a hit with identity 0.750, coverage 0.950, and completeness 0.950 scores 0.677. The second hit ranks higher because it supports most of the RNA. Distinct PDB entries are selected first, followed by backfill from the remaining chains.

The rank score is not a calibrated probability of correctness. It orders candidates without native access. Multiplication by C_q is especially useful when a high-identity local hit covers only a small part of the query, as shown by the reranking experiment in Chapter 6.

4.2.3 Coordinate transfer and gap completion

For aligned nucleotides, the template C1' coordinate is copied to the corresponding query position. Directly transferred positions retain higher confidence, whereas gaps receive lower confidence. Internal gaps are interpolated between anchors; when the anchors are too close for the number of missing steps, a perpendicular sinusoidal offset prevents the trace from collapsing onto a short line. Terminal gaps follow the nearest backbone direction, and a deterministic de novo trace provides finite coordinates when anchors are unavailable.

Curved filling makes the candidate valid before refinement; it does not reconstruct a new motif from native information. The experiment therefore bins gaps by length and measures SW-RMSD9 only at unsupported residues. Evidence of improvement is confined to gaps of 9 to 20 nucleotides.

4.3 DRfold2 branch

DRfold2 receives the RNA sequence and runs the pretrained `cfg_97` checkpoint to produce `.ret` files containing coordinates, predicted geometry, and residue-level confidence. The

pipeline averages residue confidence to rank model outputs, runs Arena to reconstruct a PDB, reads exactly one C1' atom per nucleotide, and retains the two highest-ranked valid candidates. Candidates with the wrong length or non-finite coordinates are rejected. The model is not fine-tuned on the 100-RNA cohort [16, 18].

Unlike John’s public hybrid notebook, this branch does not run Boltz or pass a Boltz structure to DRfold2 as an AF3-style restraint. DRfold2 generates direct checkpoint candidates, while TBM derives candidates from templates; the two lists meet only when the five output slots are assembled. This operational independence does not imply that pretraining is unrelated to the PDB.

The frozen implementation limits DRfold2 to 600 nucleotides. During public execution it ran on 11 of 12 targets and skipped one 720-nucleotide sequence, which retained five TBM candidates. The fallback rule was fixed before hidden evaluation.

4.4 Geometry Refinement

4.4.1 Objective

John’s TBM notebook applies a rule-based comparator before adding random diversity. It moves adjacent C1' distances toward 5.5-6.5 Å, separates clashes below 3.8 Å, and gently adjusts selected complementary bases toward 10.5 Å. The rules are applied once with confidence-dependent strength. John’s hybrid notebook also includes DRfold2 Optimization and Arena, but these belong to pretrained candidate generation rather than a common post-refiner. Geometry Refinement in this thesis operates after a complete TBM or DRfold2 C1' candidate exists.

Given source candidate $X^{(0)}$, the method finds a nearby X that reduces

$$\mathcal{L}(X) = \lambda_s \mathcal{L}_{\text{source}} + \alpha(c) \left(\lambda_b \mathcal{L}_{\text{backbone}} + \lambda_a \mathcal{L}_{\text{angle}} + \lambda_t \mathcal{L}_{\text{torsion}} + \lambda_c \mathcal{L}_{\text{clash}} + \lambda_k \mathcal{L}_{\text{kink}} + \lambda_r \mathcal{L}_{R_g} \right), \quad (4.3)$$

where c is whole-candidate confidence and $\alpha(c) = 0.2 + 0.8(1 - c)$. High-confidence candidates receive weaker geometric correction, whereas uncertain candidates can move further. The source term limits overall displacement and the remaining terms address specific local distortions.

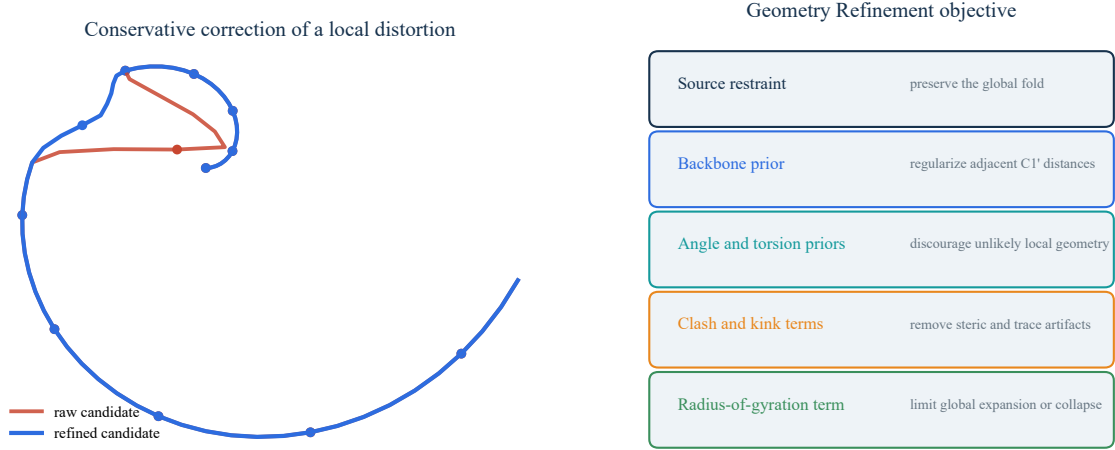


Figure 4.3: Geometry Refinement mechanism. Local distortion is regularized while a source restraint preserves the overall arrangement.

4.4.2 Geometric priors

Adjacent distance is $d_i = \|\mathbf{x}_{i+1} - \mathbf{x}_i\|_2$. On the 60 train RNAs, its mean is 6.090 Å, standard deviation is 1.110 Å, and median is 5.660 Å. The nonadjacent C1' clash threshold is 4.180 Å, and the radius-of-gyration prior is

$$\hat{R}_g(L) = 5.180L^{0.346}. \quad (4.4)$$

These values are empirical priors of the frozen project snapshot, not universal physical constants. They are estimated only on train and held fixed thereafter.

The backbone term does not use histogram likelihood. Each distance is standardized as $e_i = (d_i - \mu)/\sigma$, and Huber loss is applied to e_i . Small deviations are treated quadratically and large outliers linearly, preserving a repair gradient without allowing one long bond to dominate the optimization.

Angles over three consecutive points and torsions over four points use negative log likelihood from train histograms. Priors are separated into *pair-like* and *unpaired* contexts. A nucleotide is marked pair-like when it forms AU, UA, GC, CG, GU, or UG with a nucleotide at least four sequence positions away and their source C1' distance lies within 10.5 ± 2.5 Å. This is a native-blind geometric proxy, not an experimental base-pair assignment.

The kink guard uses a 70-degree reporting threshold and a 5-degree safety margin. It prevents a new angle from becoming sharper than 75 degrees without forcing an existing bad kink completely straight. Clash loss separates close nonadjacent points, while the R_g Huber term limits collapse and excessive expansion.

Let $h_\delta(e)$ denote Huber loss. The confidence-weighted source restraint is

$$\mathcal{L}_{\text{source}} = \frac{\sum_{i=1}^L w_i h_\delta(\|\mathbf{x}_i - \mathbf{x}_i^{(0)}\|_2)}{\sum_{i=1}^L w_i}. \quad (4.5)$$

Weights are larger for template-supported or high-confidence residues, allowing gaps and weak regions to move more than reliable scaffolds. The local confirmatory configuration uses 300 Adam steps, learning rate 0.040, gradient clipping 10.000, and source weight 1.500. The Kaggle hybrid also uses 300 steps but its production source weight is 3.000; the private score is therefore not presented as an isolated test of the exact local 1.500 configuration.

4.4.3 Baseline and output

The simple baseline contains only source restraint and backbone smoothing. John’s rule-based method supplies a second inherited comparator. Frozen results currently compare the full method with raw and simple candidates; the 20-RNA head-to-head against John’s rules is specified as an additional experiment and is not used to support a claim in version 3.

After refinement, each candidate is checked for shape, finite values, and residue order. The writer emits five coordinate triples per nucleotide in exact sample-ID order. The final pipeline contains no native-guided router, source selection, or coordinate fusion.

4.5 Algorithm summary

1. Read sequence q , temporal cutoff t_q , and excluded PDB identifiers.
2. Retrieve MMseqs2 and composite hits, then merge by chain key.
3. Remove hits that violate Equation 3.1.
4. Realign, rank, and build three TBM candidates.
5. Run DRfold2 when $L \leq 600$ and retain two candidates by model confidence.
6. Fill unavailable DRfold2 positions with TBM, maintaining five candidates.
7. Refine each candidate independently for 300 steps.
8. Validate identifiers, dimensions, uniqueness, and finite coordinates, then write CSV.

Native labels do not appear in this inference path. All pre-output decisions use sequence, template metadata, pretrained confidence, and aggregate priors. Oracle scores are

computed only after generation for analysis.

Chapter 5

Experimental Design

5.1 Testing strategy

Each research question is assigned a distinct experiment and maximum claim. RQ1 uses the 20 calibration RNAs to study retrieval sources, composite terms, reranking, and gap completion. RQ2 uses the 20 validation RNAs to compare candidate banks before refinement. RQ3 follows the full 60/20/20 protocol and opens validation only after configuration freeze. This separation prevents exploratory TBM analyses from serving as confirmatory refinement evidence.

Table 5.1: Mapping from research questions to experiments and claims.

RQ	Primary comparison	Supported claim
RQ1	MMseqs2, composite, union; component and reranking ablations	Composite expands availability; ranking effects are measured by component
RQ2	3T, 2D, union; fixed-N allocation	TBM and DRfold2 complement one another by target
RQ3	Raw, source+backbone, full refinement; cumulative and leave-one-out analyses	Refinement improves local trace within a TM preservation margin

Table 5.1 constrains interpretation before results are examined. RQ3 does not hypothesize an increase in TM-score. The frozen gate requires mean C1'-IDDT to increase, at least two of SW-RMSD9/15/31 to decrease, and TM-score loss to remain no worse than 0.005. Confidence intervals and target-level improve/regress counts are reported even when the gate passes.

5.2 Inherited comparators and comparison policy

Three John references are kept separate. The TBM-only notebook is the public method that the competition paper calls a Kaggle-winning algorithm; the hybrid notebook is a public Boltz, DRfold2, and template-fallback implementation; and 0.578 is the final score of team john. The public hybrid score of 0.576 is not treated as the exact winning submission [12, 28, 29].

On CASP15, the thesis retains an unfiltered reproduction to diagnose leakage and a time-safe reproduction for method comparison. On the principal 20-RNA cohort, a fair refinement comparator must receive exactly the same raw candidates. John’s rule-based refiner is specified as the inherited comparator; earlier CASP15 results with the project’s previous gradient refiner do not substitute for this head-to-head test.

5.3 TBM ablations

The search-source study compares MMseqs2-only, composite-only, and their union. MMseqs2 TM is conditional on returning a hit, and availability is reported alongside it. Composite and union means use all 20 targets. This avoids comparing eight easier targets with a full 20-target cohort as if the samples were equivalent.

The component study compares G , $G + L$, $G + L + F$, an equal four-term score, the full weighted score, and $\pm 25\%$ perturbations to individual weights. Useful recall at TM 0.45 is measured within a common candidate pool as an internal diagnostic. An unsafe-date variant quantifies leakage and is never considered a valid method.

The reranking study begins with identity, then adds query coverage, template completeness, and distinct-PDB selection. Gap completion compares linear and curved interpolation in length bins 1-3, 4-8, 9-20, and above 20 nucleotides. Gaps within a target are averaged before target-level bootstrap.

5.4 Complementarity analysis

Source candidates are evaluated before refinement. For target i , let T_i contain the three TBM scores and D_i the two DRfold2 scores. The source and union ceilings are

$$O_T(i) = \max T_i, \quad O_D(i) = \max D_i, \quad O_U(i) = \max(T_i \cup D_i). \quad (5.1)$$

The union gain $O_U - O_T$ measures the available value of two additional DRfold2 models. Because the union has more candidates, it combines source diversity and count effects.

Fixed- N comparisons control this issue. At $N = 2$, $2T$ is compared with $1T + 1D$; at $N = 3$, $3T$ is compared with $2T + 1D$. Candidates are ordered by blind source confidence. The cache contains only three TBM and two DRfold2 models per target, so a complete five-slot sweep from $5T$ to $5D$ is not identifiable without duplicating candidates.

5.5 Geometry-refinement analysis

Train estimates priors only. On calibration, cumulative ablation adds source, backbone, angle, torsion, clash, kink, and R_g terms, while leave-one-out analysis removes each term from the full objective. The selected configuration satisfies local improvement and TM preservation, then is serialized before validation.

Validation compares raw candidates, simple source-plus-backbone smoothing, and the full objective. Primary local metrics are C1'-IDDT, SW-RMSD9, and SW-RMSD15. Best-of-five TM is the global safeguard, with whole-structure C1 RMSD and SW-RMSD31 as secondary measures. Backbone deviation, clash, sharp kink, angle NLL, torsion NLL, and R_g error are mechanism diagnostics because they lie in or near the objective.

John's rule-based refiner will be evaluated on the same raw candidates with the same fixed native conformation. Until that result is available, version 3 does not claim that Geometry Refinement outperforms John's refiner.

A fixed- $N = 3$ factorial forms four cells: $3T$ raw, $3T$ refined, $2T + 1D$ raw, and $2T + 1D$ refined. It estimates source-bank gain with refinement off, refinement effect within each bank, and the interaction. This design directly tests whether refinement can explain the hybrid TM increase.

5.6 External evaluation and reproducibility

Two thesis submissions have scores: 0.602 private for TBM and 0.614 for the hybrid. They establish deployment performance but are not a one-variable ablation, because the TBM submission uses an earlier baseline refiner while the hybrid changes both candidate bank and refinement. Their 0.012 difference is therefore a complete-pipeline effect.

A preregistered private factorial will generate four outputs from one candidate run: $5T$ raw, $5T$ refined, $3T+2D$ raw, and $3T+2D$ refined. Version 3 does not fabricate these

unavailable scores and does not assign the existing Kaggle gain separately to DRfold2 or refinement.

Repository scripts generate all local tables. Bootstrap seed is 2025, and frozen configurations, prior provenance, and candidate caches accompany the artifacts. Audits require one metadata row per target, no cross-split groups, exactly three TBM and two DRfold2 candidates per target, finite coordinates, and a fixed reference conformation between raw and refined evaluation.

Chapter 6

Results

6.1 RQ1: TBM retrieval and ranking

6.1.1 Retrieval sources

MMseqs2 found a template for 8 of 20 calibration RNAs, an availability of 40%. On those eight targets, top-1 TM was 0.660 and best-of-five TM was 0.678. Composite search produced candidates for all 20 targets but reached a lower whole-cohort mean of 0.376. The union maintained full availability and increased the mean to 0.390.

Table 6.1: Retrieval-source results on 20 calibration RNAs. MMseqs2 quality is conditional on the eight targets with a hit.

Source	Targets with hits	Availability	Best-of-five TM when available
MMseqs2-only	8/20	0.40	0.678
Composite-only	20/20	1.00	0.376
MMseqs2 + composite	20/20	1.00	0.390

The sources have distinct roles. MMseqs2 candidates are strong when a homolog is recognized, while composite search fills the remaining 12 targets at lower average precision. The union gained 0.014 over composite-only, although its interval $[-0.004, 0.043]$ included zero. The clearest benefit is therefore availability, not uniform superiority of composite hits.

6.1.2 Composite components

The full weighted score reached 0.376, whereas global alignment alone reached 0.381. The G-only advantage was 0.005 with confidence interval $[-0.002, 0.015]$. Perturbing individual

weights by $\pm 25\%$ produced scores from 0.375 to 0.377. The heuristic is locally stable but does not clearly outperform global alignment on this cohort.

The $G + L + F$ variant nevertheless had useful recall 0.676 compared with 0.635 for G-only within the common hit pool. This pattern is consistent with a recall heuristic: additional signals introduce distinct candidates even when mean oracle TM does not rise clearly. Useful recall remains an internal diagnostic rather than a replacement for TM.

Removing the temporal filter increased mean TM from 0.376 to 0.403, a change of 0.027. The interval $[-0.000, 0.071]$ was wide and 15 targets were unchanged. This is not evidence for an unsafe method; it shows that a few leaked templates can shift the cohort mean.

6.1.3 Reranking and gap completion

Adding query coverage to identity increased top-five TM from 0.372 to 0.388, a gain of 0.016 with interval $[-0.004, 0.045]$. Completeness and distinct-PDB selection did not change oracle TM on this cohort, although the latter increased the mean number of distinct PDB sources. Coverage is therefore the dominant ranking signal in Table 6.2, while later rules primarily support robustness.

Table 6.2: TBM reranking ablation.

Ranking	Top-1 TM	Top-3 TM	Top-5 TM
Identity only	0.334	0.364	0.372
Identity $\times$ coverage	0.364	0.378	0.388
+ template completeness	0.364	0.378	0.388
+ distinct PDB	0.364	0.378	0.388

Curved gap filling showed clear improvement only for gaps of 9 to 20 nucleotides. It reduced SW-RMSD9 by 0.037 Å with interval $[0.009, 0.068]$. Bins 1-3 and 4-8 showed no clear difference from linear filling, and gaps above 20 were unchanged. The curve has enough span to avoid a straight bridge in the middle bin but insufficient information to reconstruct very long missing motifs.

6.1.4 Comparison with the inherited TBM pipeline

On the same 12 CASP15 RNAs under the same time-safe condition, the local reproduction of John’s TBM notebook reached 0.298, whereas the thesis TBM pipeline reached 0.307. The

absolute difference was 0.009, and the thesis pipeline performed better on 9 of 12 RNAs. This comparison supports the combined TBM changes, including dual retrieval, reranking, and candidate handling. It does not assign the full difference to any one change; the preceding ablations provide the component-level evidence.

6.2 RQ2: Source complementarity

Before refinement, the best of three TBM candidates reached a mean TM-score of 0.565, whereas the best of two DRfold2 candidates reached 0.503. Keeping all five candidates increased the mean to 0.588. The absolute gain over TBM was 0.023, with a 95% confidence interval of $[0.006, 0.048]$. TBM supplied the stronger candidate for 12 of 20 RNAs, while DRfold2 won on the other eight. The pretrained model was therefore useful because it covered cases missed by the template bank, not because it had a higher average score.

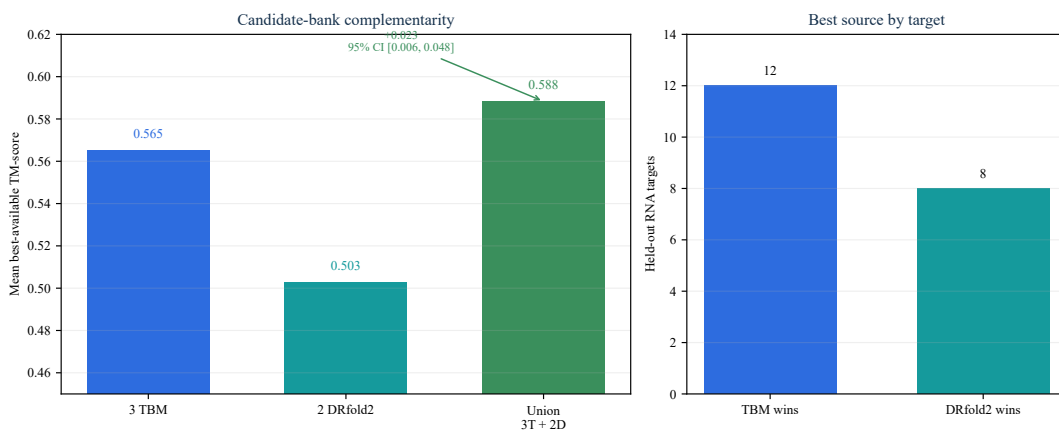


Figure 6.1: Candidate-bank complementarity on 20 validation RNAs. The union interval is obtained by target-level bootstrap.

The DRfold2 source mean was 0.063 below TBM. Moreover, the best of five candidates has a mechanical advantage over the best of three because it has two additional chances to contain a good structure. Fixed-size comparisons were therefore required to separate source diversity from candidate-count advantage.

With two prediction slots, $1T + 1D$ exceeded $2T$ by 0.031, with interval $[0.003, 0.068]$. With three slots, $2T + 1D$ exceeded $3T$ by 0.016, although the interval $[-0.002, 0.041]$ included zero and ten RNAs tied. The strongest evidence is therefore the two-slot comparison. Once three TBM candidates already provide some diversity, the added value of replacing one of them with DRfold2 becomes smaller and less certain.

Table 6.3: Candidate allocation with Geometry Refinement disabled.

Comparison	Baseline mean	Hybrid mean	Mean gain and 95% CI
$2T$ versus $1T + 1D$	0.549	0.580	+0.031 [0.003, 0.068]
$3T$ versus $2T + 1D$	0.565	0.581	+0.016 [−0.002, 0.041]
$3T$ versus $3T + 2D$	0.565	0.588	+0.023 [0.006, 0.048]

Table 6.3 distinguishes fixed-size and production-bank effects. In both the two-slot and three-slot comparisons, replacing one TBM candidate with DRfold2 increased the mean, but only the two-slot confidence interval lay entirely above zero.

6.3 RQ3: Geometry Refinement

6.3.1 Confirmatory results

Full refinement increased C1'-lDDT from 0.660 to 0.666, an absolute gain of 0.007 with interval [0.001, 0.013]; 12 of 20 RNAs improved. SW-RMSD9 decreased from 2.651 to 2.611 Å, a 0.040 Å error reduction with interval [0.029, 0.052]; 19 of 20 RNAs improved. SW-RMSD15 decreased by 0.017 Å, with interval [0.001, 0.031]. These three results consistently indicate a short-range effect.

Global TM-score changed from 0.588 to 0.587, or −0.001, with interval [−0.003, 0.000]. The unrounded interval contained zero, and the mean change remained inside the predeclared 0.005 preservation margin. Whole-structure C1 RMSD and SW-RMSD31 showed no clear change. Refinement therefore passed the local-quality gate but did not provide evidence of a better global fold.

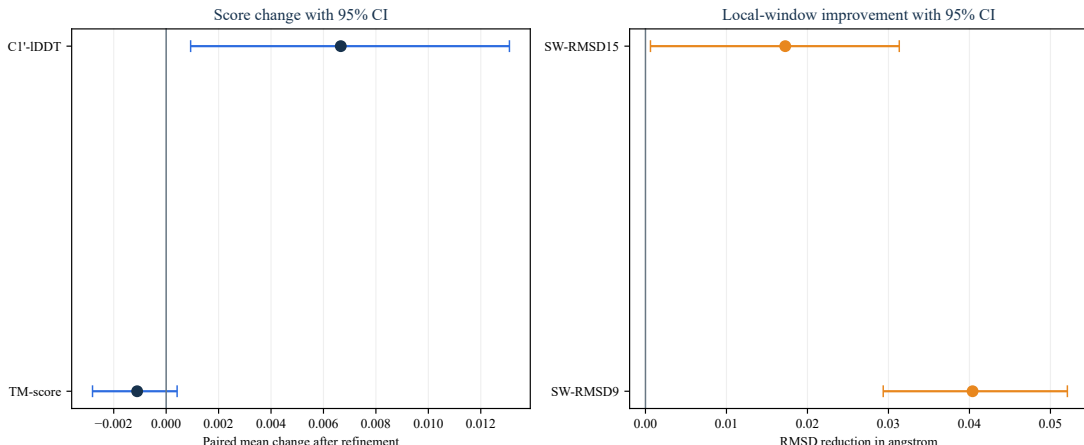


Figure 6.2: Confirmatory geometry-refinement effects. Score changes and RMSD reductions use separate axes and units.

Table 6.4: Raw and full Geometry Refinement on 20 validation RNAs.

Metric	Raw	Refined	Improvement	95% CI
Best-of-five TM	0.588	0.587	-0.001	$[-0.003, 0.000]$
C1'-IDDT	0.660	0.666	+0.007	$[0.001, 0.013]$
SW-RMSD9 (Å)	2.651	2.611	+0.040	$[0.029, 0.052]$
SW-RMSD15 (Å)	4.273	4.256	+0.017	$[0.001, 0.031]$
Backbone deviation (Å)	1.040	0.786	+0.254	diagnostic
Clash per residue	0.065	0.023	+0.042	diagnostic

The dominant pattern in Table 6.4 is stronger improvement at shorter spatial scales. SW-RMSD9 is the most consistent effect, SW-RMSD15 is smaller, and SW-RMSD31 and global RMSD are inconclusive. This scale gradient matches the design: refinement adjusts neighboring relationships rather than rearranging the fold.

6.3.2 Simple baseline and ablation

Source-plus-backbone smoothing increased IDDT by 0.008, close to the full method’s gain of 0.007. The direct full-minus-simple difference was -0.001, with interval $[-0.003, 0.001]$, so the full objective did not outperform the simple baseline on IDDT. However, the full method reduced SW-RMSD9 by a further 0.031 Å, with interval $[0.020, 0.044]$, and avoided worsening the angle, torsion, clash, and kink diagnostics.

Leave-one-out analysis linked terms to mechanisms. Removing backbone nearly eliminated the IDDT gain; removing angle or torsion degraded the corresponding NLL; removing

kink increased sharp bends; and removing clash or R_g worsened clash diagnostics. These results show that the objective behaves as designed, but optimized diagnostics are not treated as independent evidence. The primary claim remains based on lDDT and sliding-window RMSD.

6.3.3 Source-by-refinement factorial

Within the $3T$ bank, refinement changed TM by -0.006 , with interval $[-0.015, 0.001]$. Within $2T + 1D$, the effect was -0.003 , with interval $[-0.005, 0.000]$. The interaction was $+0.003$, with an interval containing zero. Table 6.5 therefore shows a positive source-bank effect before refinement and no positive TM effect from refinement in either bank.

Table 6.5: Fixed- $N = 3$ factorial separating source composition and Geometry Refinement.

Candidate bank	Raw mean TM	Refined mean TM
$3T + 0D$	0.565	0.559
$2T + 1D$	0.581	0.579

The factorial contradicts a simple explanation in which refinement causes the hybrid global-TM gain. Replacing one of three TBM candidates with DRfold2 already produced a 0.016 mean increase with refinement disabled. Refinement slightly reduced TM in both banks. Source diversity is therefore the better-supported global mechanism, while refinement acts on local quality.

6.3.4 Subgroups and failure cases

In a post hoc stratification, initially low-quality candidates gained 0.016 lDDT with interval $[0.009, 0.024]$. High-quality candidates changed by -0.002 , with interval $[-0.006, 0.002]$. This pattern motivates a future blind quality gate but cannot revise the current method because the groups were examined after validation.

Three failures remain visible. Targets 9E2Z_F and 8K7W_A lost about 0.009 TM, while 9B0S_Et lost about 0.011 lDDT. Conservative refinement does not imply monotonic improvement for every target, and both uncertainty and failure counts are needed beside the mean.

6.4 External Kaggle evaluation

The thesis TBM pipeline scored 0.601 public and 0.602 private. The final pipeline reached 0.628 public and 0.614 private, producing absolute differences of 0.027 and 0.012 between the two thesis submissions. Against public reference points, the final pipeline was 0.021 above John’s TBM-only notebook, 0.038 above John’s hybrid notebook, and 0.036 above the winning team’s private score [28, 29]. Table 6.6 keeps these reference points separate because a public notebook version is not identical to the final winning submission.

Table 6.6: Kaggle comparison. A hyphen indicates that the source did not report that score.

Method	Public TM	Private TM
John’s hybrid notebook	-	0.576
Winning team john	-	0.578
John’s TBM-only notebook	-	0.593
Thesis TBM pipeline	0.601	0.602
Thesis final pipeline	0.628	0.614

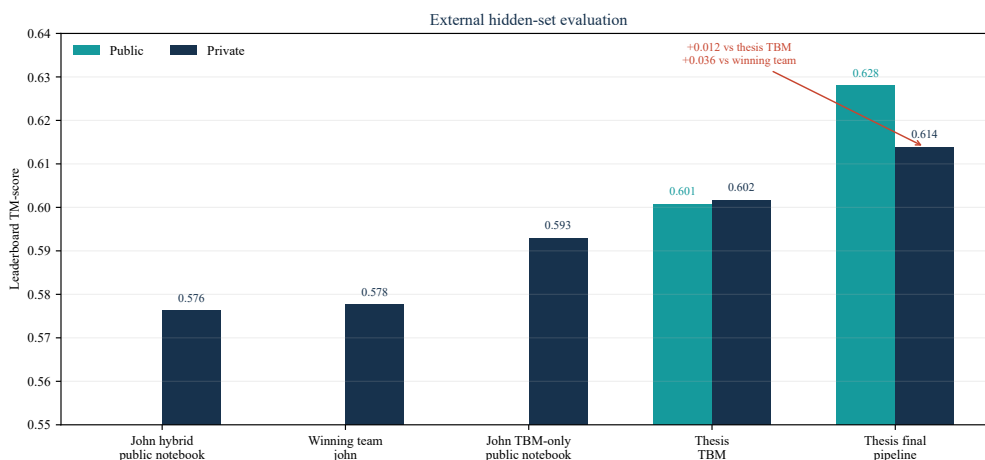


Figure 6.3: Evaluation on hidden structures. The three John reference points report only the published private score; the two thesis pipelines report both public and private scores.

The public execution provides deployment checks. DRfold2 generated two candidates for 11 of 12 targets, the 720-nucleotide target used five TBM candidates, and refinement completed for all 60 structures. Output shape and identifiers were valid, and the run did not read native labels. These observations support execution integrity but cannot identify which hidden targets produced the score increase.

The Kaggle result must be read with the local factorial. The final submission both replaced two TBM slots and applied refinement, making the private change a pipeline-level

effect. Local evidence shows that hybrid composition improves global TM before refinement, whereas refinement improves local metrics and slightly reduces TM. Together, the two evidence levels explain the higher score without attributing it wholesale to Geometry Refinement.

Chapter 7

Discussion

7.1 Inheritance and thesis-specific changes

The pipeline was not developed from scratch. It inherits the winning solution’s central sequence: retrieve known structures with TBM, generate additional candidates with a pre-trained model, postprocess the coordinates, and submit five structures. Stating this provenance is necessary because a leaderboard difference is scientifically interpretable only when the reused and modified components are separated.

The thesis retains the “retrieve and transfer” principle of John’s TBM method but adds two complementary retrieval sources, a merged template snapshot, strict temporal filtering, query-coverage reranking, distinct-PDB selection, and curved completion for medium gaps. On the same time-safe CASP15 set, John’s local TBM reproduction scored 0.298 and the thesis TBM scored 0.307. The absolute difference was 0.009, approximately 3.0% relative to 0.298, with the thesis method winning on 9 of 12 RNAs. This is a pipeline-level TBM comparison; the RQ1 ablations provide the evidence for individual changes.

The pretrained branch also follows John’s hybrid idea but does not reproduce every branch of the public notebook. The thesis uses a fixed DRfold2 to Arena to C1’ path and does not use Boltz. John’s final rule-based adjustment and the thesis Geometry Refinement also differ: John’s rules make one pass over adjacent distances, clashes, and selected base pairs, whereas the thesis optimizes a multi-term objective over several steps. A same-candidate head-to-head comparison has not yet been completed, so this draft does not claim that the thesis refiner outperforms John’s refiner.

7.2 Why the hybrid pipeline has value

The gain over the TBM-only reference does not require one source to dominate everywhere. TBM had the higher source mean and won on 12 targets, while DRfold2 won on the remaining eight. Under best-of-five scoring, a distinct source only needs to win a substantial subset of targets to improve the union mean. Complementarity is therefore a target-wise property rather than a comparison of source averages.

The two-slot fixed-N result strengthens this argument by removing candidate-count advantage. Replacing one of two TBM structures with a DRfold2 structure retained a positive confidence interval. The three-slot effect was weaker because three TBM models already contained more diversity and the validation cohort contained only 20 targets. The three-plus-two allocation is consequently a reasonable compromise for the hidden distribution, not a universal optimum.

Competition data provide a second explanation. The analysis reported that 19 of 20 private targets had a prior structure with TM-align above 0.45, a higher template-rich fraction than CASP15 [12]. A large template snapshot and broad retrieval were advantageous in this regime. The same paper cautions that this ordering may not generalize to template-poor RNA [12]; DRfold2 remains an important hedge against that part of the distribution.

7.3 Why refinement does not increase TM-score

The source restraint places the optimum near the starting candidate. Backbone, angle, torsion, and clash terms operate mainly on short-range relationships and contain no signal for changing topology or moving an entire domain. Improvement in lDDT and SW-RMSD9 with nearly unchanged TM is therefore consistent with the intended mechanism.

An incorrect global fold requires long-range contacts, motif information, or a different template. Local optimization around one source cannot invent this evidence. Nucleic-acid refinement studies likewise distinguish improved stereochemistry from systematic native RMSD improvement [20]. Geometry Refinement should therefore be interpreted as trace regularization, with source diversity setting the primary global ceiling.

The simple baseline matched the full method on lDDT, identifying backbone smoothing as the principal mechanism for that score. The full objective showed clearer value for SW-RMSD9 and balanced angle, torsion, clash, and kink diagnostics. This narrows the contribution: the full objective is not necessary for every local gain, but it provides a more balanced trace without disrupting the average fold.

7.4 Interpreting the leaderboard

The 0.614 private score is measured on hidden structures. It exceeded the thesis TBM pipeline by 0.012, or approximately 2.0% relative to 0.602. It also exceeded John’s TBM-only notebook by 0.021, John’s public hybrid notebook by 0.038, and the winning team score by 0.036. The absolute differences are the primary comparisons; the corresponding relative differences are approximately 3.5%, 6.5%, and 6.3%.

These values establish that the complete implementation transfers beyond the local cohort, but a scalar leaderboard score does not provide a per-target counterfactual. The two thesis submissions changed both candidate composition and postprocessing. In addition, the locally confirmed Geometry Refinement used source weight 1.5, whereas the Kaggle run used 3.0. The 0.012 private difference is therefore a complete-pipeline effect, not a one-variable measurement of DRfold2 or of the exact local refinement configuration.

A defensible explanation has three layers. The integrity audit shows that the submission does not use native labels and enforces temporal filtering. The local complementarity analysis identifies a positive global mechanism. The local refinement analysis identifies a separate local mechanism and shows that it does not increase TM. Together, these layers explain performance while preserving the distinction between external validity and component causality.

Distribution shift is also visible. The frozen hybrid rule scored 0.465 on the 12 CASP15 targets and was worse than the local TBM bank, whereas the complete Kaggle pipeline improved private TM by 0.012. This reversal shows that effective allocation depends on the target distribution. Reporting both outcomes is more credible than selecting the favorable cohort and argues against a universally optimal mixture.

7.5 Threats to validity

7.5.1 Internal validity

Calibration and validation contain only 20 targets each, leaving wide intervals for several ablations. Target bootstrap handles dependence among candidates but cannot create information beyond the observed RNAs [26]. Multiple subgroup analyses also increase the chance of incidental patterns, so they are labeled post hoc and do not change the current method.

The cache contains only three TBM and two DRfold2 candidates. Fixed-N comparisons up to three positions are identified, but a full five-position allocation sweep is not.

Future work should generate at least five independent candidates from each source before opening a new cohort.

7.5.2 External validity

Local validation covers single-chain RNA up to 96 nucleotides. Refinement results do not directly generalize to 720-nucleotide RNA, multichain complexes, or RNA-protein interfaces. Such systems have different interaction patterns and experimental uncertainties [1, 5].

The Kaggle private set is unusually template rich [12]. Its score supports the competition distribution but does not guarantee the same ordering on a template-poor benchmark. A new temporal benchmark released after 2026 would provide a stronger generalization test.

7.5.3 Metric validity

TM-score and C1'-IDDT evaluate a coarse-grained trace. They do not directly measure base orientation, hydrogen bonding, ion coordination, or full-atom stereochemistry, all of which contribute to tertiary interactions and RNA function [4, 2]. A high-scoring trace is therefore not yet a complete model for docking or mechanistic interpretation.

Angle NLL, torsion NLL, clash, and kink are objective diagnostics and favor the full method by construction. They explain mechanism but cannot replace independent metrics. Prioritizing IDDT, sliding-window RMSD, and TM reduces circularity, while full-atom validation remains necessary for biochemical use.

7.6 Future work

The first priority is a same-candidate comparison between John's rule-based refiner and the thesis Geometry Refinement. The second is the locked private 2×2 experiment: 5T raw, 5T refined, 3T+2D raw, and 3T+2D refined. These experiments determine whether the thesis can claim an improvement over inherited postprocessing or only a complete-pipeline gain.

The next extension is quality-gated refinement. Post hoc results suggest larger IDDT gains for weak candidates than strong ones. A blind gate could skip refinement when source confidence and self-consistency are high, but it must be preregistered on a new calibration set and tested on an unopened cohort.

A second direction is greater within-source diversity. Five independent TBM and five DRfold2 models would support a complete allocation sweep and measure the marginal return

of each position. A third direction is full-atom reconstruction with an evaluated tool such as Arena, followed by stereochemical metrics designed for RNA [21, 8].

Finally, temporal metadata should fail closed, and self-PDB exclusion should parse target reference identifiers directly. These changes improve auditability rather than score. In a field with continuously released structural data, reconstructing what was known at prediction time is essential to meaningful evaluation.

Chapter 8

Conclusion

This thesis inherited the winning solution’s TBM-plus-pretrained framework and rebuilt it as a time-safe, component-tested pipeline with a separate Geometry Refinement stage. Rather than treating the final score as the only evidence, the study separated retrieval, source diversity, and local postprocessing into three questions with distinct ablations.

For RQ1, MMseqs2 produced strong candidates when a homolog was found but covered only 8 of 20 calibration RNAs. Composite search expanded availability to all 20 targets. The full weighted score did not clearly beat global alignment alone, and query coverage was the most influential reranking component. Composite search is therefore supported as a recall heuristic within time-safe TBM.

For RQ2, TBM had higher average source quality, but TBM and DRfold2 won on different targets. Their union improved mean best-available TM by 0.023 with a positive confidence interval, and the fixed- $N = 2$ comparison retained a 0.031 gain. These results directly show that source diversity can improve best-of-five performance without one source universally outperforming the other.

For RQ3, Geometry Refinement increased C1’-IDDT by 0.007, reduced SW-RMSD9 by 0.040 Å, and changed TM by -0.001 . A simple smoother explained much of the IDDT gain, while the full objective improved short-window RMSD and geometric diagnostics. The supported conclusion is local trace correction with average global-fold preservation.

The final pipeline reached 0.628 public and 0.614 private. Its private score was 0.021 above John’s TBM-only notebook, 0.038 above John’s public hybrid notebook, and 0.036 above the winning team. These remain complete-pipeline comparisons, not evidence that every component individually outperforms its inherited counterpart. The local factorial identifies candidate complementarity as the better-supported global mechanism, while Geometry Refinement provides a separate local benefit. The same-candidate refiner comparison and

private 2×2 factorial remain necessary before the thesis is frozen.

Appendix A

Reproducibility Artifacts

The principal artifacts are:

- `scripts/run_tbm_whitebox_ablation.py` for the TBM component study;
- `scripts/summarize_hybrid_complementarity.py` for source allocation;
- `reports/thesis_notes/tbm_whitebox_ablation.md` for RQ1 tables;
- `reports/thesis_notes/hybrid_complementarity.md` for RQ2 tables;
- `reports/thesis_notes/kaggle_submission_analysis.md` for external scores and manifests;
- `scripts/generate_thesis_v3_figures.py` for all data figures in this draft.

The confirmatory runner, its detailed RQ3 report, and the frozen configuration remain in the repository. The final release should add a manifest with the Git commit, Python environment, template-database checksum, pretrained-model version, and exact command for every table.

Appendix B

Pre-submission Checklist

All data figures are generated from frozen aggregates by the script listed in Appendix A. Before submission, each plotted value must be checked against its source table, together with grayscale contrast and text size at 100%. If a structure overlay is added, it must use a public validation RNA selected by a predeclared rule and must report the target, candidate source, before-and-after metrics, and generation script. A hidden Kaggle structure must not be used as an illustrative example.

Bibliography

- [1] J. Zhang, Y. Fei, L. Sun, and Q. C. Zhang, “Advances and opportunities in RNA structure experimental determination and computational modeling,” *Nature Methods*, vol. 19, pp. 1193-1207, 2022. doi: 10.1038/s41592-022-01623-y.
- [2] X. Cao, Y. Zhang, Y. Ding, and Y. Wan, “Identification of RNA structures and their roles in RNA functions,” *Nature Reviews Molecular Cell Biology*, vol. 25, pp. 784-801, 2024. doi: 10.1038/s41580-024-00748-6.
- [3] I. Tinoco Jr. and C. Bustamante, “How RNA folds,” *Journal of Molecular Biology*, vol. 293, no. 2, pp. 271-281, 1999. doi: 10.1006/jmbi.1999.3001.
- [4] S. E. Butcher and A. M. Pyle, “The Molecular Interactions That Stabilize RNA Tertiary Structure: RNA Motifs, Patterns, and Networks,” *Accounts of Chemical Research*, vol. 44, no. 12, pp. 1302-1311, 2011. doi: 10.1021/ar200098t.
- [5] H. Ma, X. Jia, K. Zhang, and Z. Su, “Cryo-EM advances in RNA structure determination,” *Signal Transduction and Targeted Therapy*, vol. 7, article 58, 2022. doi: 10.1038/s41392-022-00916-0.
- [6] S. K. Burley and coauthors, “Protein Data Bank: the single global archive for 3D macromolecular structure data,” *Nucleic Acids Research*, vol. 47, no. D1, pp. D520-D528, 2019. doi: 10.1093/nar/gky949.
- [7] J. A. Cruz and coauthors, “RNA-Puzzles: A CASP-like evaluation of RNA three-dimensional structure prediction,” *RNA*, vol. 18, no. 4, pp. 610-625, 2012. doi: 10.1261/rna.031054.111.
- [8] M. Magnus, M. Antczak, T. Zok, and coauthors, “RNA-Puzzles toolkit: a computational resource of RNA 3D structure benchmark datasets, structure manipulation, and evaluation tools,” *Nucleic Acids Research*, vol. 48, no. 2, pp. 576-588, 2020. doi: 10.1093/nar/gkz1108.

- [9] R. Das and coauthors, “Assessment of three-dimensional RNA structure prediction in CASP15,” *Proteins: Structure, Function, and Bioinformatics*, vol. 91, no. 12, pp. 1747-1770, 2023. doi: 10.1002/prot.26602.
- [10] C. Bernard, G. Postic, S. Ghannay, and F. Tahi, “State-of-the-RNArt: benchmarking current methods for RNA 3D structure prediction,” *NAR Genomics and Bioinformatics*, vol. 6, no. 2, article lqae048, 2024. doi: 10.1093/nargab/lqae048.
- [11] X. Wang, S. Yu, E. Lou, Y. L. Tan, and Z. J. Tan, “RNA 3D Structure Prediction: Progress and Perspective,” *Molecules*, vol. 28, no. 14, article 5532, 2023. doi: 10.3390/molecules28145532.
- [12] Y. Lee, S. He, T. Oda, and coauthors, “Template-based RNA structure prediction advanced through a blind code competition,” *bioRxiv*, preprint 2025.12.30.696949, 2025. doi: 10.64898/2025.12.30.696949.
- [13] M. Steinegger and J. Söding, “MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets,” *Nature Biotechnology*, vol. 35, pp. 1026-1028, 2017. doi: 10.1038/nbt.3988.
- [14] S. B. Needleman and C. D. Wunsch, “A general method applicable to the search for similarities in the amino acid sequence of two proteins,” *Journal of Molecular Biology*, vol. 48, no. 3, pp. 443-453, 1970. doi: 10.1016/0022-2836(70)90057-4.
- [15] T. F. Smith and M. S. Waterman, “Identification of common molecular subsequences,” *Journal of Molecular Biology*, vol. 147, no. 1, pp. 195-197, 1981. doi: 10.1016/0022-2836(81)90087-5.
- [16] Y. Li, C. Zhang, C. Feng, R. Pearce, P. L. Freddolino, and Y. Zhang, “Integrating end-to-end learning with deep geometrical potentials for ab initio RNA structure prediction,” *Nature Communications*, vol. 14, article 5745, 2023. doi: 10.1038/s41467-023-41303-9.
- [17] T. Shen, Z. Hu, S. Sun, and coauthors, “Accurate RNA 3D structure prediction using a language model-based deep learning approach,” *Nature Methods*, vol. 21, pp. 2287-2298, 2024. doi: 10.1038/s41592-024-02487-0.
- [18] Y. Li, C. Feng, X. Zhang, S. Tsukiyama, D. Feng, and Y. Zhang, “DRfold2 is a deep learning-based tool that enables efficient and accurate RNA structure prediction,” *PLOS Biology*, vol. 24, no. 2, article e3003659, 2026. doi: 10.1371/journal.pbio.3003659.

- [19] R. J. L. Townshend, S. Eismann, A. M. Watkins, and coauthors, “Geometric deep learning of RNA structure,” *Science*, vol. 373, no. 6558, pp. 1047-1051, 2021. doi: 10.1126/science.abe5650.
- [20] P. Stasiewicz, S. Mukherjee, C. Nithin, and J. M. Bujnicki, “QRNAS: software tool for refinement of nucleic acid structures,” *BMC Structural Biology*, vol. 19, article 5, 2019. doi: 10.1186/s12900-019-0103-1.
- [21] Z. R. Perry, A. M. Pyle, and C. Zhang, “Arena: Rapid and Accurate Reconstruction of Full Atomic RNA Structures From Coarse-grained Models,” *Journal of Molecular Biology*, vol. 435, article 168210, 2023. doi: 10.1016/j.jmb.2023.168210.
- [22] Y. Zhang and J. Skolnick, “Scoring function for automated assessment of protein structure template quality,” *Proteins: Structure, Function, and Bioinformatics*, vol. 57, no. 4, pp. 702-710, 2004. doi: 10.1002/prot.20264.
- [23] C. Zhang, M. Shine, A. M. Pyle, and Y. Zhang, “US-align: universal structure alignments of proteins, nucleic acids, and macromolecular complexes,” *Nature Methods*, vol. 19, pp. 1109-1115, 2022. doi: 10.1038/s41592-022-01585-1.
- [24] V. Mariani, M. Biasini, A. Barbato, and T. Schwede, “lDDT: a local superposition-free score for comparing protein structures and models using distance difference tests,” *Bioinformatics*, vol. 29, no. 21, pp. 2722-2728, 2013. doi: 10.1093/bioinformatics/btt473.
- [25] W. Kabsch, “A solution for the best rotation to relate two sets of vectors,” *Acta Crystallographica Section A*, vol. 32, pp. 922-923, 1976. doi: 10.1107/S0567739476001873.
- [26] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman and Hall/CRC, 1993.
- [27] Stanford University and Kaggle, “Stanford RNA 3D Folding,” Kaggle competition documentation, 2025. Available: <https://www.kaggle.com/competitions/stanford-rna-3d-folding>.
- [28] G. John Rao (JaeJohn), “RNA 3D folds: TBM-only approach,” Kaggle notebook, version 1, 2025. Available: <https://www.kaggle.com/code/jaejohn/rna-3d-folds-tbm-only-approach>.
- [29] G. John Rao (JaeJohn), “[sub-1] 4/4 hybrid - final take,” Kaggle notebook, version 4, 2025. Available: <https://www.kaggle.com/code/jaejohn/sub-1-4-4-hybrid-final-take>.